

Corrected Assignment 4 Prompt Log

Prompts transcribed from the attached screenshots, followed by the matching screenshot in sequence

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1. Prism project skeleton

Create a clean LaTeX article skeleton for a scientific research paper. Include title, authors, affiliations, abstract, keywords, Introduction, Background and Literature Review, Main Thematic Discussion, Current Approaches, Research Challenges and Limitations, Future Directions, Conclusion, and References. Do not generate scientific content.

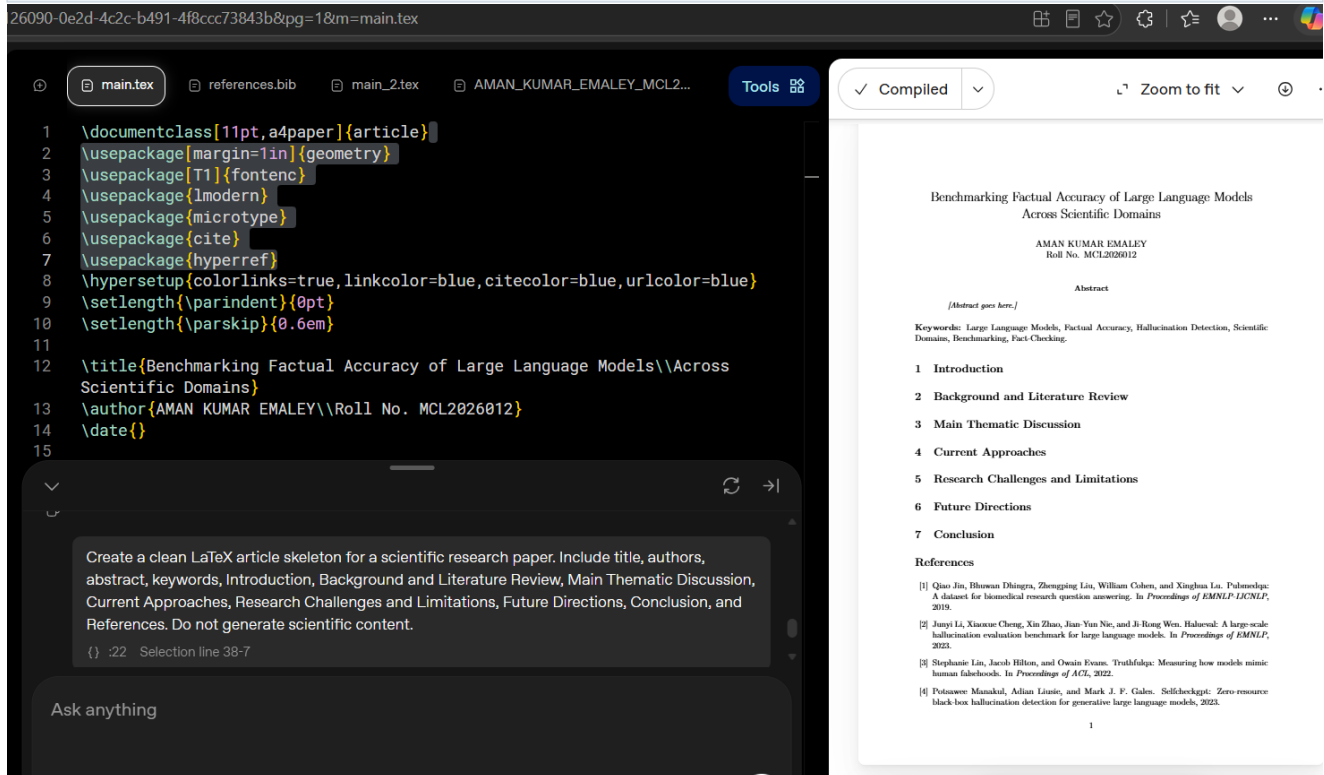


Figure 1. Matching evidence screenshot from the attached materials.

2. Title, abstract, and keywords conversion

Convert the following text into clean LaTeX. Preserve the exact wording, terminology, scientific meaning, numerical values, and citations. Do not add, remove, or reinterpret any scientific claim.

Title: Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains

Author: AI Researcher, Gemini Research Institute, August 2026

Abstract: The rapid integration of Large Language Models (LLMs) into scientific research, education, and clinical practice necessitates rigorous evaluation of their factual accuracy. Unlike general-purpose conversational agents, models deployed in scientific domains—such as medicine, physics, and biology—must adhere to strict standards of truthfulness and reliability. This paper reviews the current landscape of benchmarking factual accuracy in LLMs across various scientific disciplines. We examine prominent datasets, including MedQA, SciBench, and ScienceQA, and discuss the persistent challenge of "hallucinations." Furthermore, we analyze current approaches to mitigate these errors, such as Retrieval-Augmented Generation (RAG) and Chain-of-Thought (CoT) prompting, while highlighting research gaps and future directions for developing dynamic, domain-specific evaluation frameworks.

Keywords: Large Language Models, Factual Accuracy, Hallucination, Scientific Benchmarking, Artificial Intelligence.

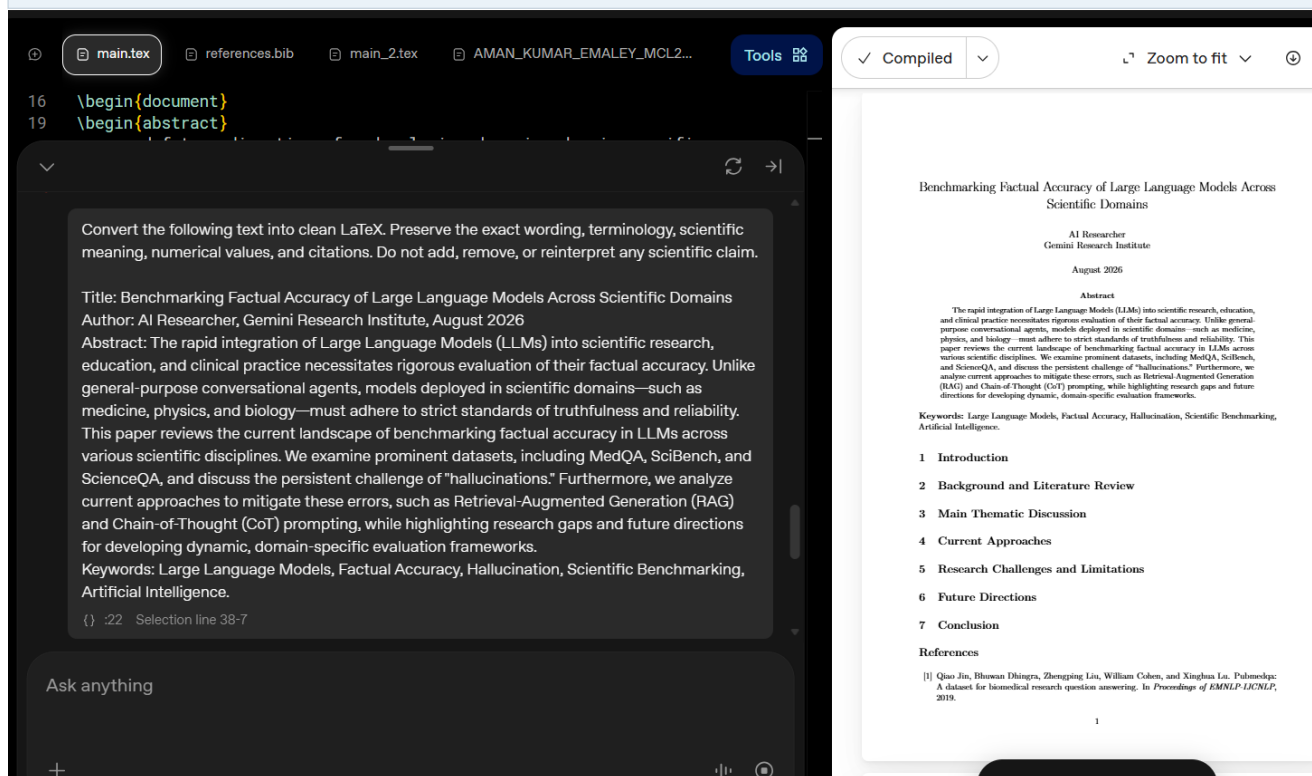


Figure 2. Matching evidence screenshot from the attached materials.

3. Structured section conversion — literature review

Convert the following sections into properly structured LaTeX using appropriate section commands. Preserve paragraph order, citations, terminology, and all technical details. Do not summarize and do not introduce new content. Replace the bracketed citations (like [1,2]) with proper LaTeX citation commands (like `\cite{...}`) if you are ready to map them, otherwise leave them as bracketed numbers for now.

The screenshot displays a LaTeX editor interface with a file explorer at the top showing `main.tex`, `references.bib`, `main_2.tex`, and `AMAN_KUMAR_EMALEY_MCL2...`. The main editor area shows LaTeX code for a document structure:

```
16 \begin{document}
25 \section{Introduction}\label{sec:introduction}
```

A prompt is entered in the editor: "Convert the following sections into properly structured LaTeX using appropriate section commands. Preserve paragraph order, citations, terminology, and all technical details. Do not summarize and do not introduce new content. Replace the bracketed citations (like [1,2]) with proper LaTeX citation commands (like \cite{...}) if you are ready to map them, otherwise leave them as bracketed numbers for now."

The output of the conversion is shown in a preview window on the right, titled "Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains" by "AI Researcher" from "Gemini Research Institute", dated "August 2026".

Abstract

The rapid integration of Large Language Models (LLMs) into scientific research, education, and clinical practice necessitates rigorous evaluation of their factual accuracy. Unlike general-purpose conversational agents, models deployed in scientific domains—such as medicine, physics, and biology—must adhere to strict standards of truthfulness and reliability. This paper reviews the current landscape of benchmarking factual accuracy in LLMs across various scientific disciplines. We examine prominent datasets, including MedQA, SciBench, and ScienceQA, and discuss the persistent challenge of "hallucination." Furthermore, we analyze current approaches to mitigate these errors, such as Retrieval-Augmented Generation (RAG) and Chain-of-Thought (CoT) prompting, while highlighting research gaps and future directions for developing dynamic, domain-specific evaluation frameworks.

Keywords: Large Language Models, Factual Accuracy, Hallucination, Scientific Benchmarking, Artificial Intelligence.

1 Introduction

Large Language Models (LLMs), driven by the transformer architecture, have demonstrated unprecedented capabilities in natural language understanding and generation [1,2]. As these models increasingly penetrate scientific domains, their utility is heavily contingent upon their factual accuracy. In fields like biomedicine, a fabricated claim or "hallucination" can have severe real-world consequences [3]. Benchmarking these models across scientific domains requires specialized datasets that test complex reasoning rather than mere rote memorization. This paper explores the methodologies and benchmarks currently used to assess the factual accuracy of LLMs in science.

2 Background and Literature Review

The evaluation of language models has historically relied on generalized benchmarks like GLUE and SuperGLUE. However, as models like GPT-4 and PaLM evolved, their ability to bypass these benchmarks necessitated more rigorous, domain-specific tests [4]. Hallucination—defined as the generation of text that is fluent but factually incorrect or ungrounded—remains a fundamental flaw in autoregressive models [5]. To quantify this, Liu et al. introduced TruthfulQA, a benchmark specifically designed to measure how models mimic human falsehoods and misconceptions [6]. Hendrycks et al. expanded on this by proposing the Massive Multitask Language Understanding

Figure 3. Matching evidence screenshot from the attached materials.

4. Structured section conversion — domain benchmarking

Convert the following sections into properly structured LaTeX using appropriate section commands. Preserve paragraph order, citations, terminology, and all technical details. Do not summarize and do not introduce new content. Replace the bracketed citations (like [7]) with proper LaTeX citation commands (like `\cite{...}`) if you are ready to map them, otherwise leave them as bracketed numbers for now.

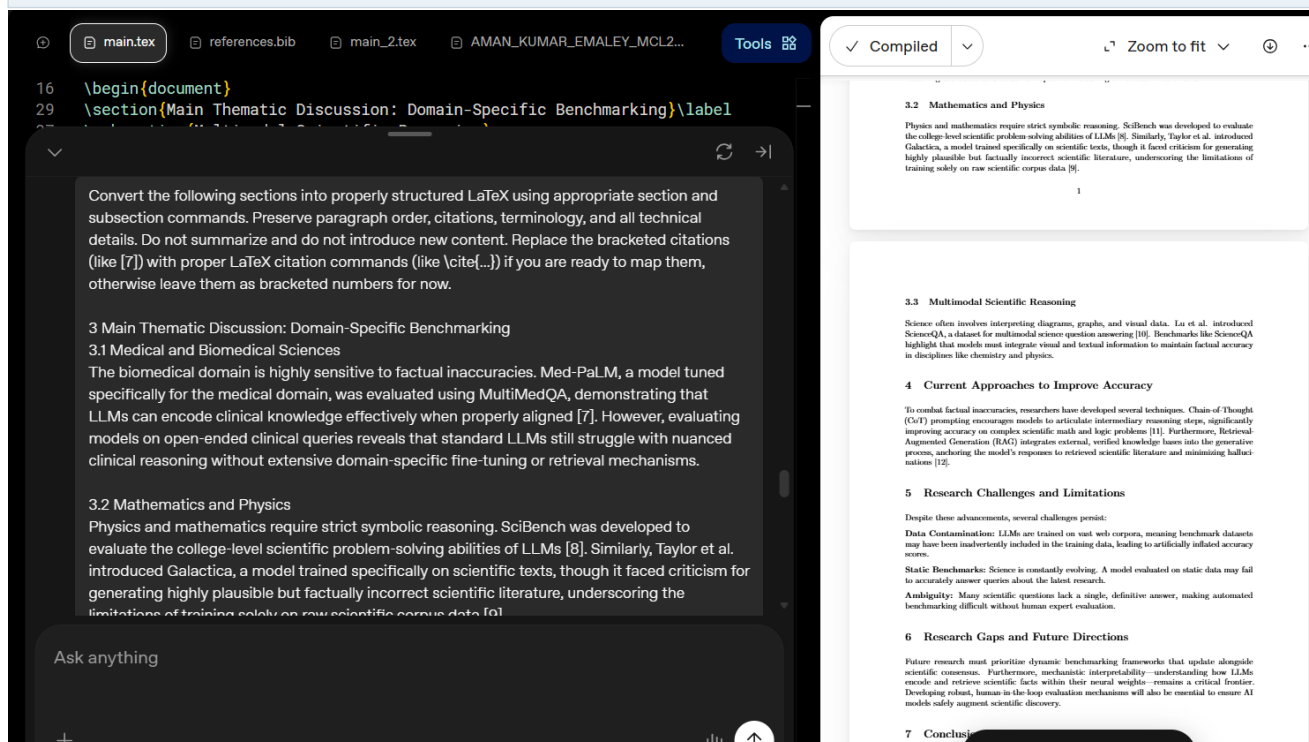


Figure 4. Matching evidence screenshot from the attached materials.

5. Table insertion

Create and insert the following table into my paper under Section 3, "Domain-Specific Benchmarking", immediately after the introductory paragraph of that section.

Important formatting requirements:

- Use a standard LaTeX table environment.
- Keep the table centered and completely inside the A4 text width.
- Use `\small` font.
- Use `\booktabs` (`\toprule`, `\midrule`, `\bottomrule`).
- Use `\tabularx` so the table automatically fits `\linewidth`.
- Do not create excessive vertical whitespace before or after the table.
- Keep row spacing compact but readable.
- Do not allow the table to overflow the page margins.
- Caption must be above the table.
- Label must be exactly: `tab:benchmarks`
- Caption: "Representative resources for evaluating factuality and scientific reasoning in LLMs."

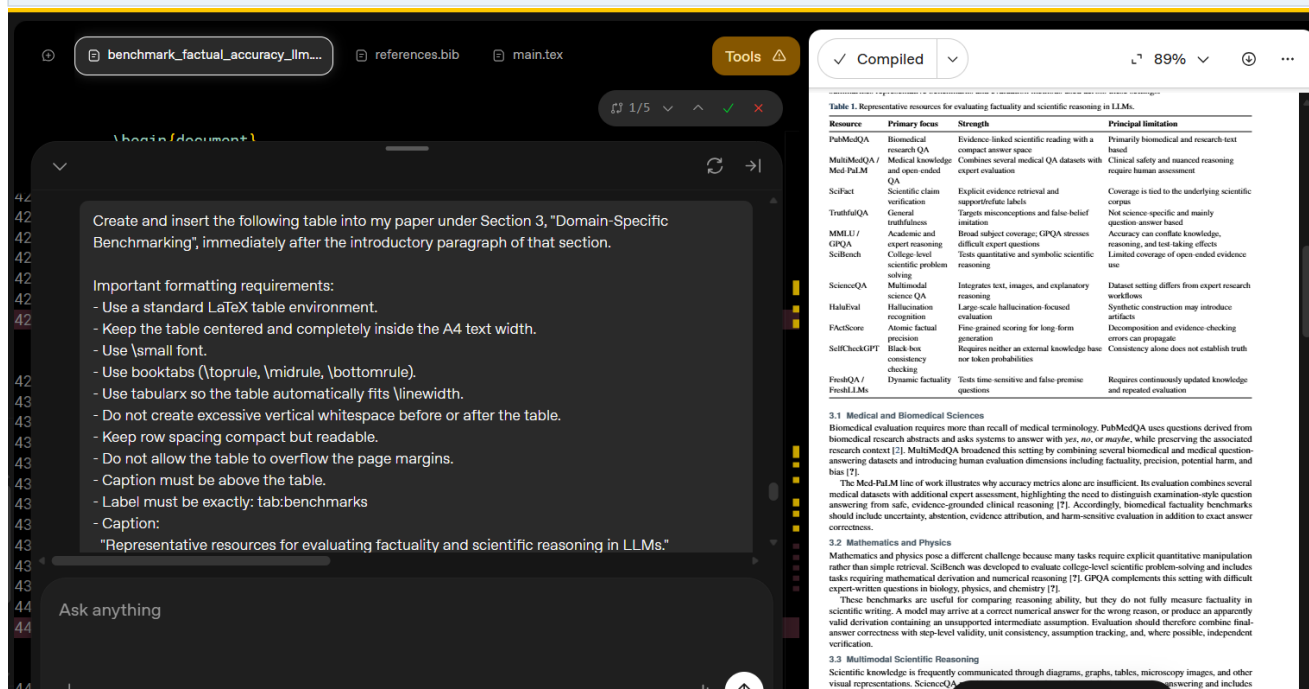


Figure 5. Matching evidence screenshot from the attached materials.

6. Figure creation and placement

Please generate the LaTeX code to insert a TikZ figure representing the "Cross-Domain Scientific LLM Factualty Benchmark".

Placement requirements:

- Insert the figure exactly at the end of Section 4.3 ("Consistency, Atomic Scoring, and Dynamic Evaluation").
- It must be placed immediately after the sentence: "These developments motivate a unified benchmark protocol in which evaluation mode and factuality metrics are reported separately, as shown in Figure 1".
- It must appear immediately before Section 5 ("Research Challenges and Limitations").
- Use the [htbp] or [H] (with \usepackage{float}) float specifier to ensure it stays in this exact location.

Formatting requirements:

- Keep the figure centered horizontally.
- Keep the entire figure inside the A4 \linewidth.
- Do not allow any node, arrow, text, or box to cross the page margins.

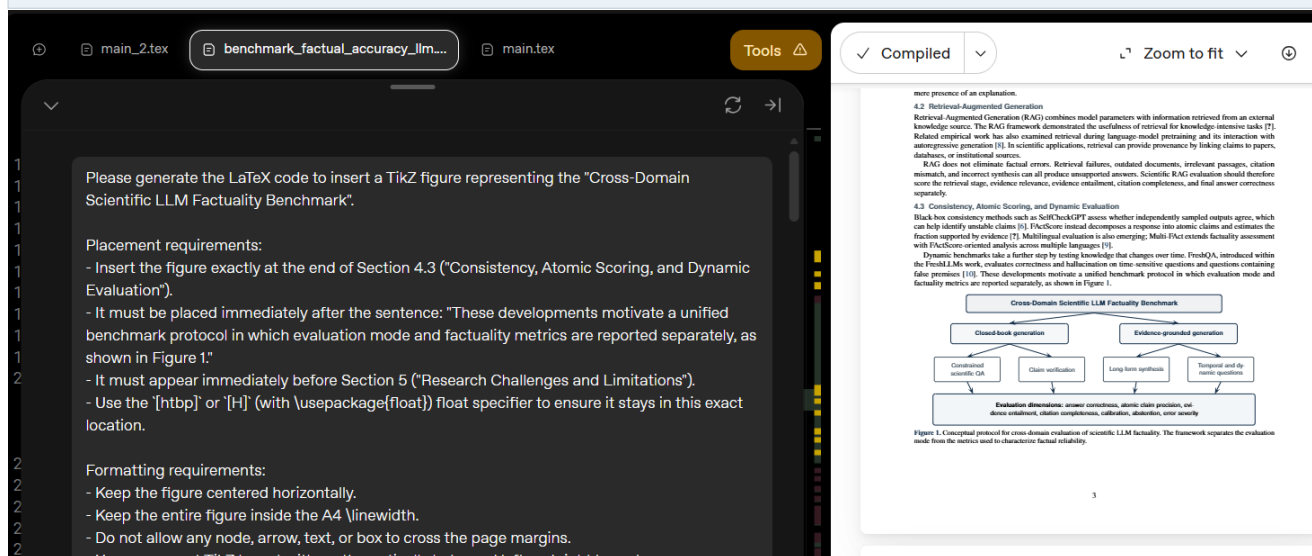


Figure 6. Matching evidence screenshot from the attached materials.

7. Overleaf portability implementation

Implementation according to Assignment 4, Part L:

The exported project was organized with `main.tex` as the root file, `references.bib` as the bibliography, and the `Figures/` folder preserved with the project evidence. In Overleaf, the project was imported, `main.tex` was selected as the main document, and the PDF preview was checked for the title, section structure, bibliography, table, figure, citations, and cross-references. The original paper contains no paper-specific mathematical equation, so the equation requirement is recorded as not applicable. The screenshot below documents the compiled Overleaf project state.

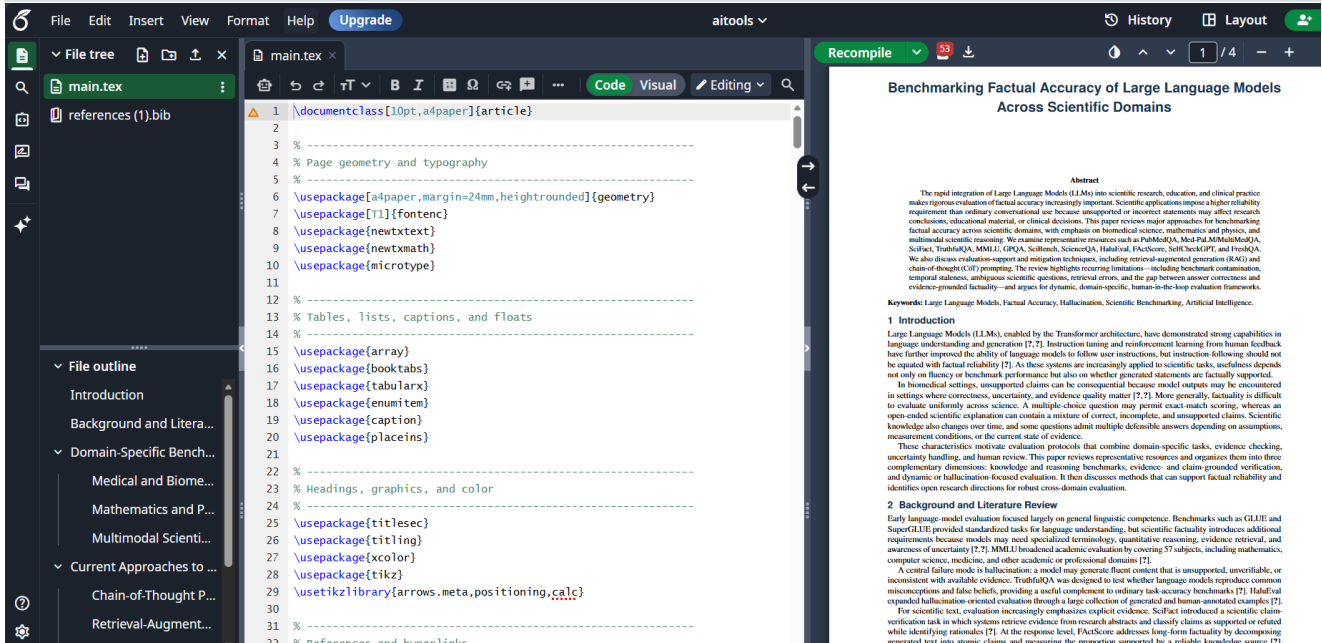


Figure 7. Matching evidence screenshot from the attached materials.